NVMe-over-TCP ≈ NVMe-over-RDMA:

CPU-efficient Remote Storage Access with i10



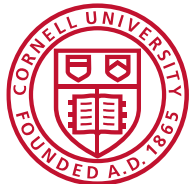
Jaehyun Hwang



Qizhe Cai

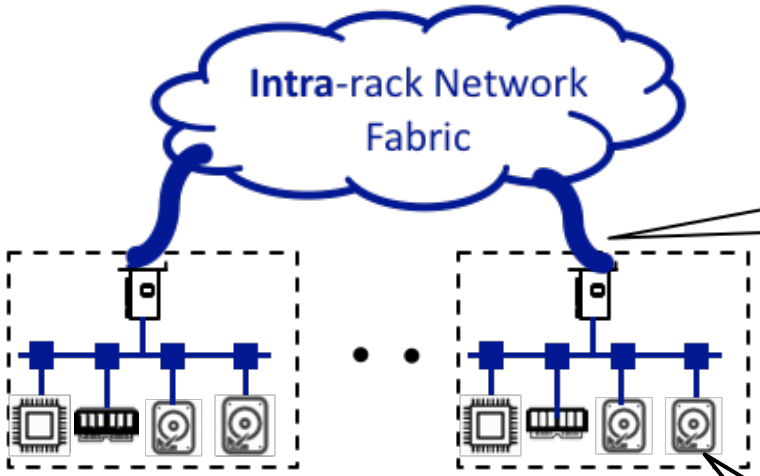


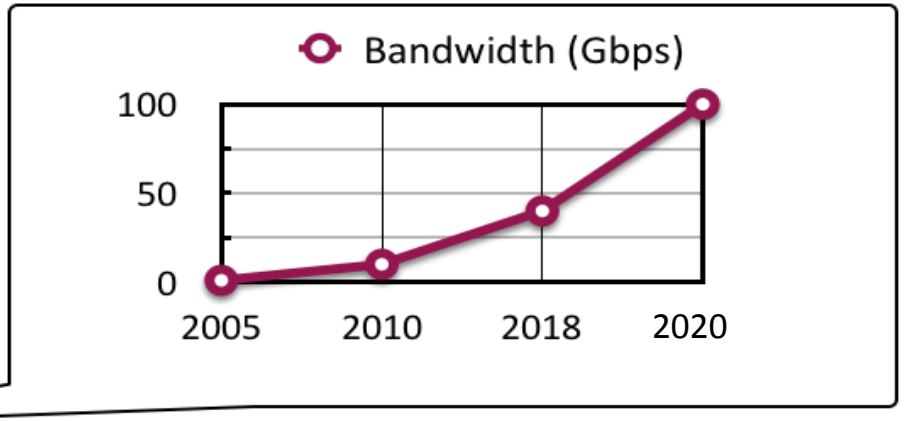
Rachit Agarwal



Cornell University

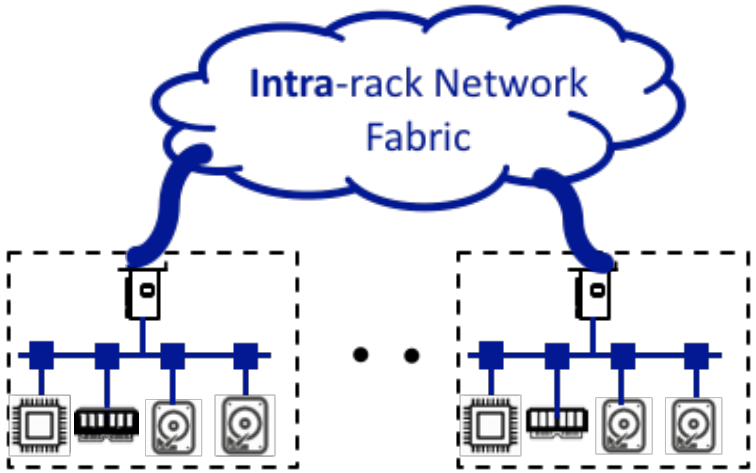
Two hardware trends





- **Emergence of NVMe SSDs:**
  - **>1M IOPS (read), >400K IOPS (write)**

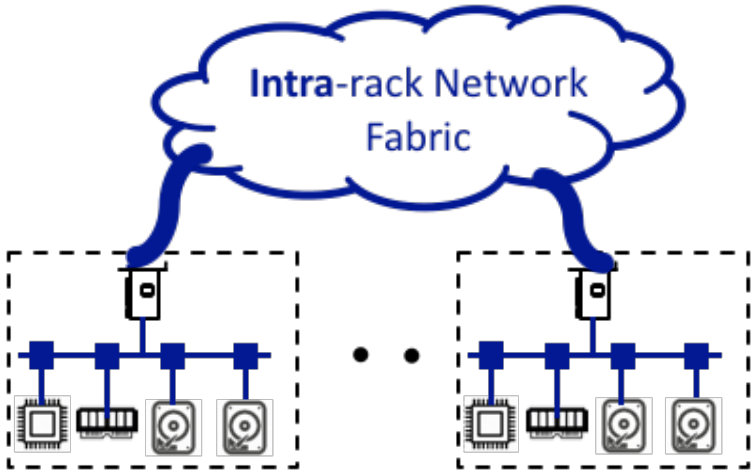
Two hardware trends

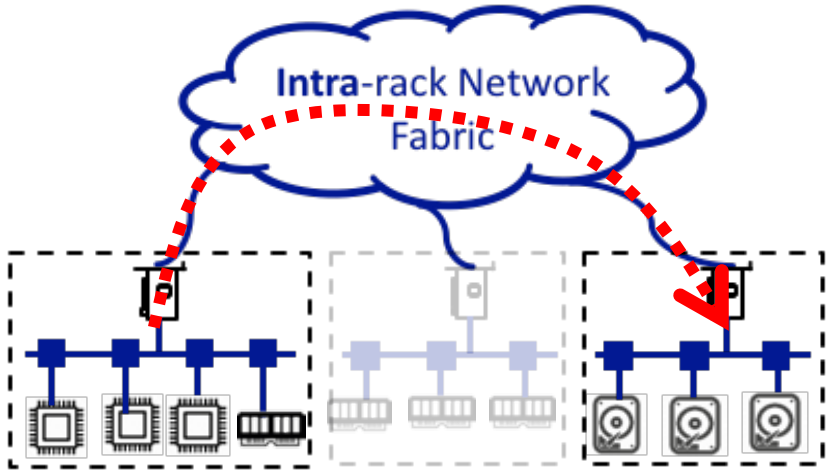


1. High-throughput NVMe SSD, high-bandwidth access links

Bottlenecks pushed back to OS, software stack!

Two hardware trends

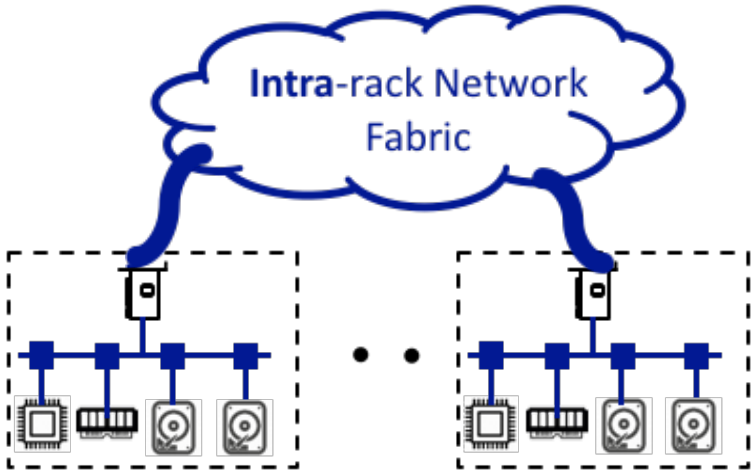


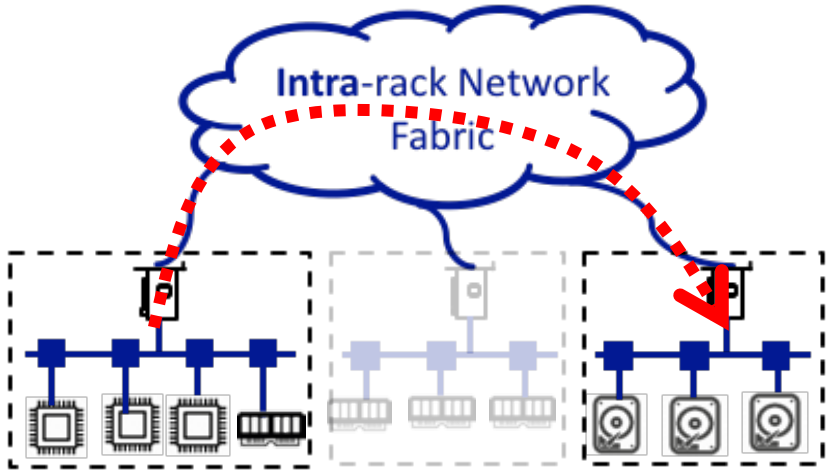


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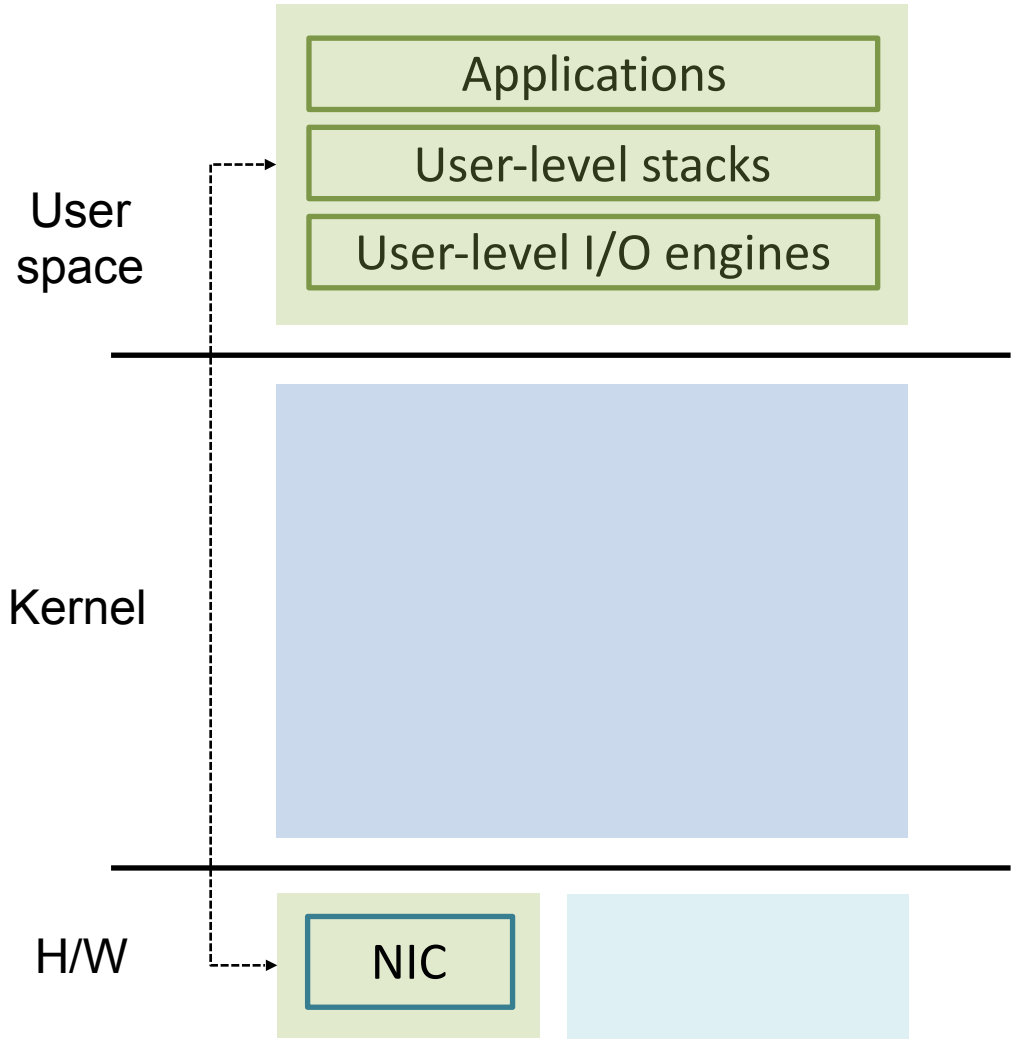
1. High-throughput NVMe SSD, high-bandwidth access links

2. Apps use disaggregated storage via Remote I/O

Bottlenecks pushed back to OS, software stack!

Increasing overlap between storage and network data paths!

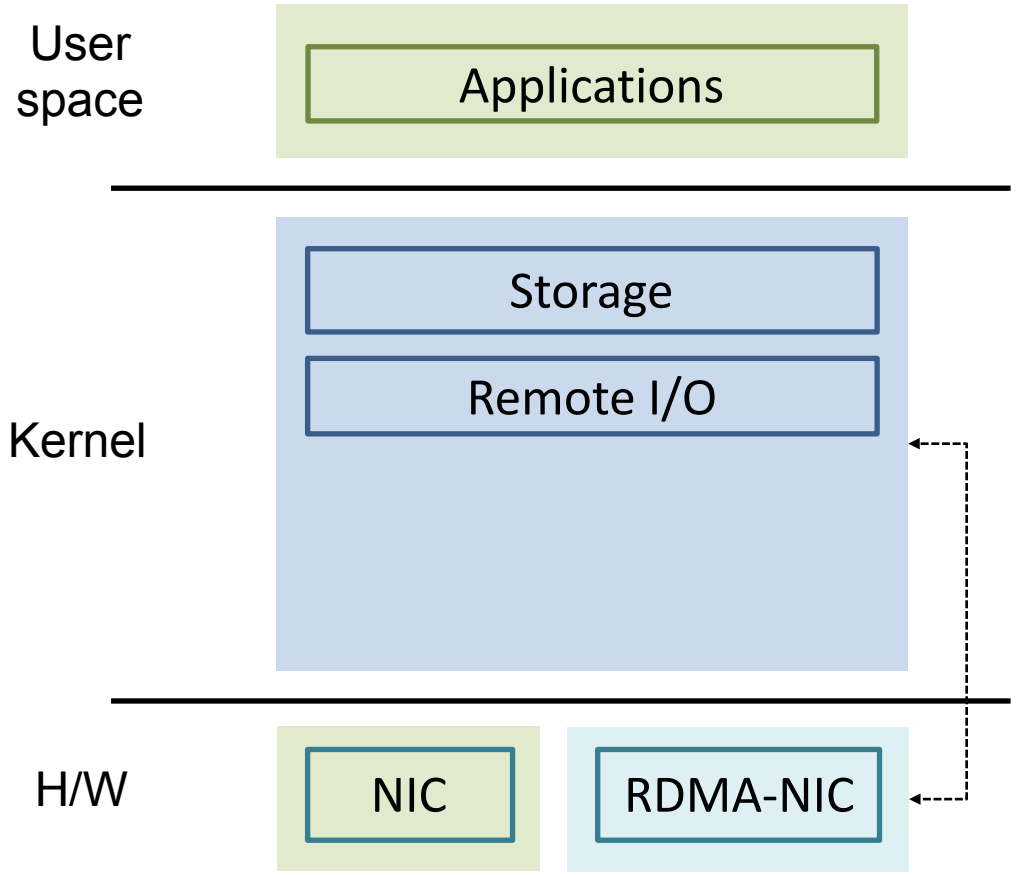
Existing approaches for remote I/O



User-space storage and network stacks

- Storage + Remote I/O (user) + DPDK
- Performance: Good!

Existing approaches for remote I/O



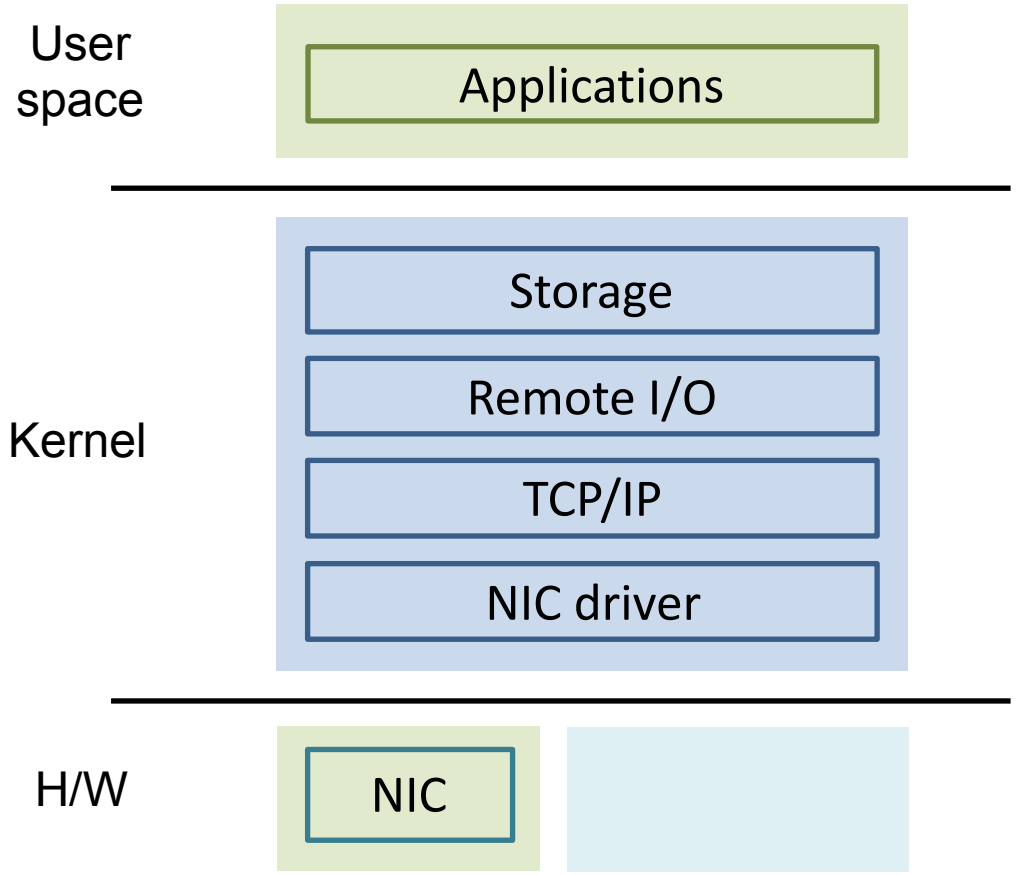
User-space storage and network stacks

- Storage + Remote I/O (user) + DPDK
- Performance: Good!

NVMe-over-RDMA

- Storage + Remote I/O (kernel) + RDMA
- Performance: Good!

Existing approaches for remote I/O



User-space storage and network stacks

- Storage + Remote I/O (user) + DPDK
- Performance: Good!

In-Kernel (NVMe-over-TCP)

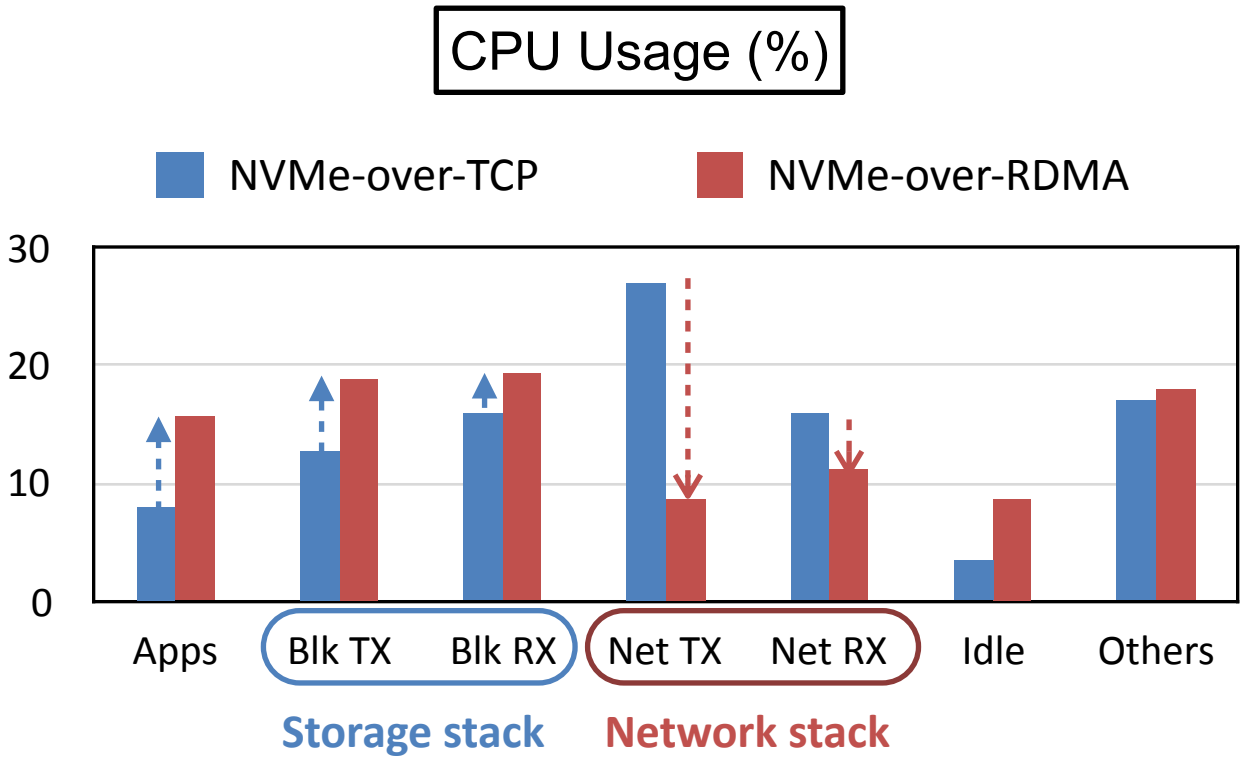
- Storage + Remote I/O + TCP (all in the kernel)
- Performance: Not-so-good (~2-3x worse)!

Fundamental?

NVMe-over-RDMA

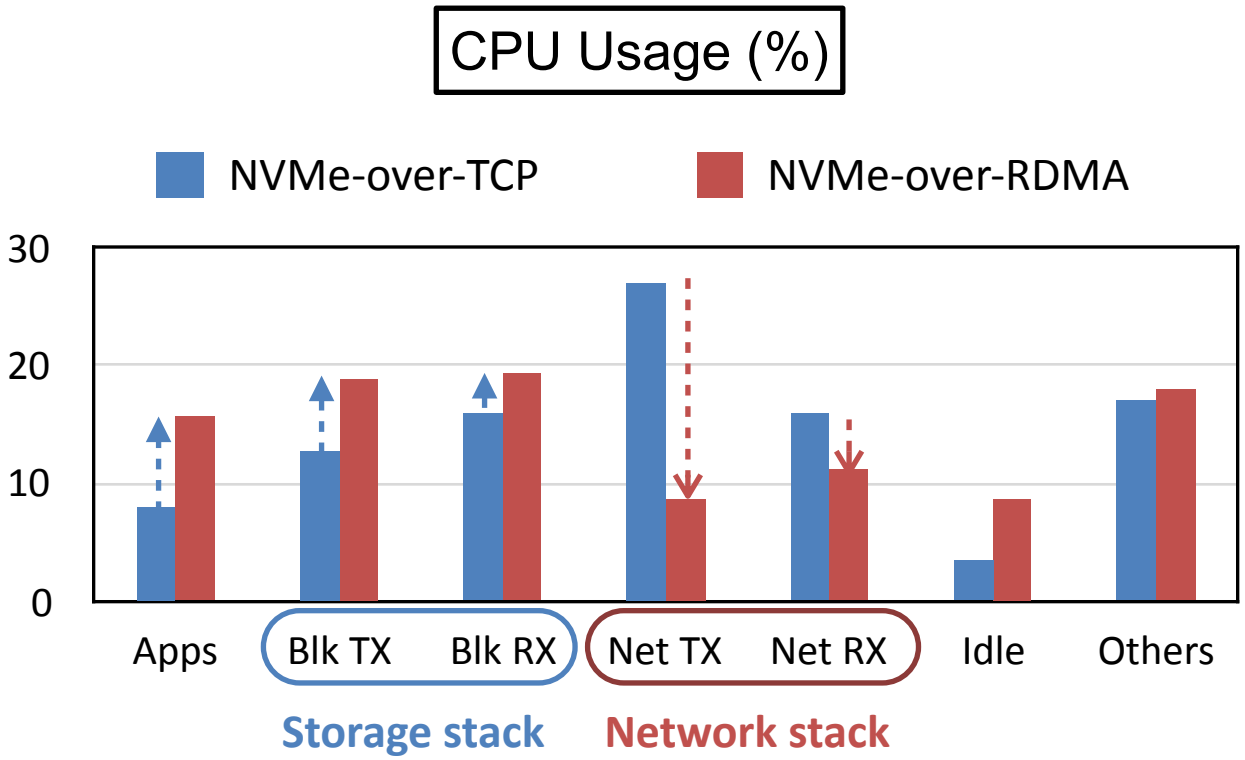
- Storage + Remote I/O (kernel) + RDMA
- Performance: Good!

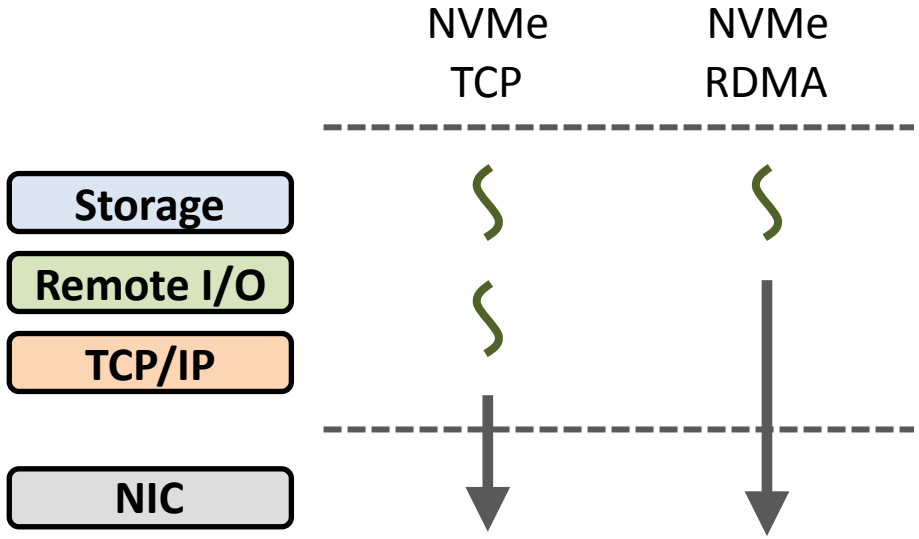
Remote Storage Access Overheads: TCP vs. RDMA



Network processing overhead!

Remote Storage Access Overheads: TCP vs. RDMA



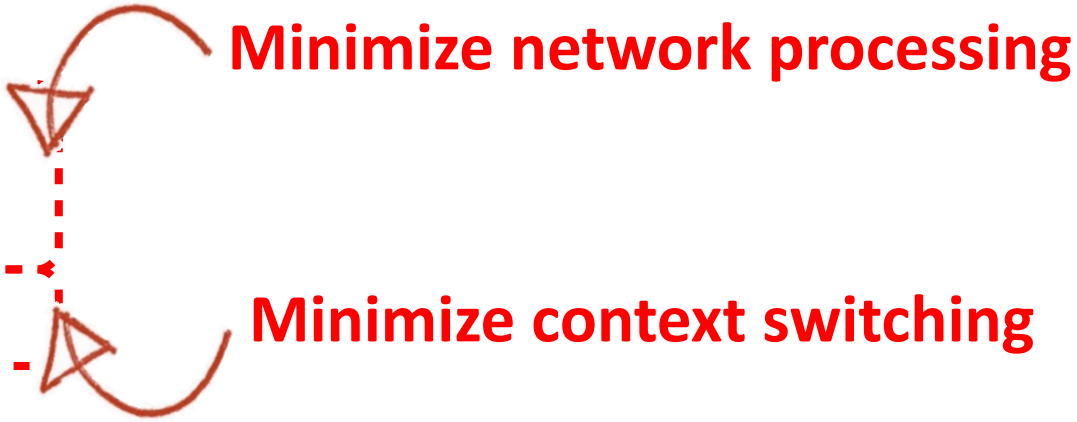


Network processing overhead!

Context switching overhead!

i10: key take-away

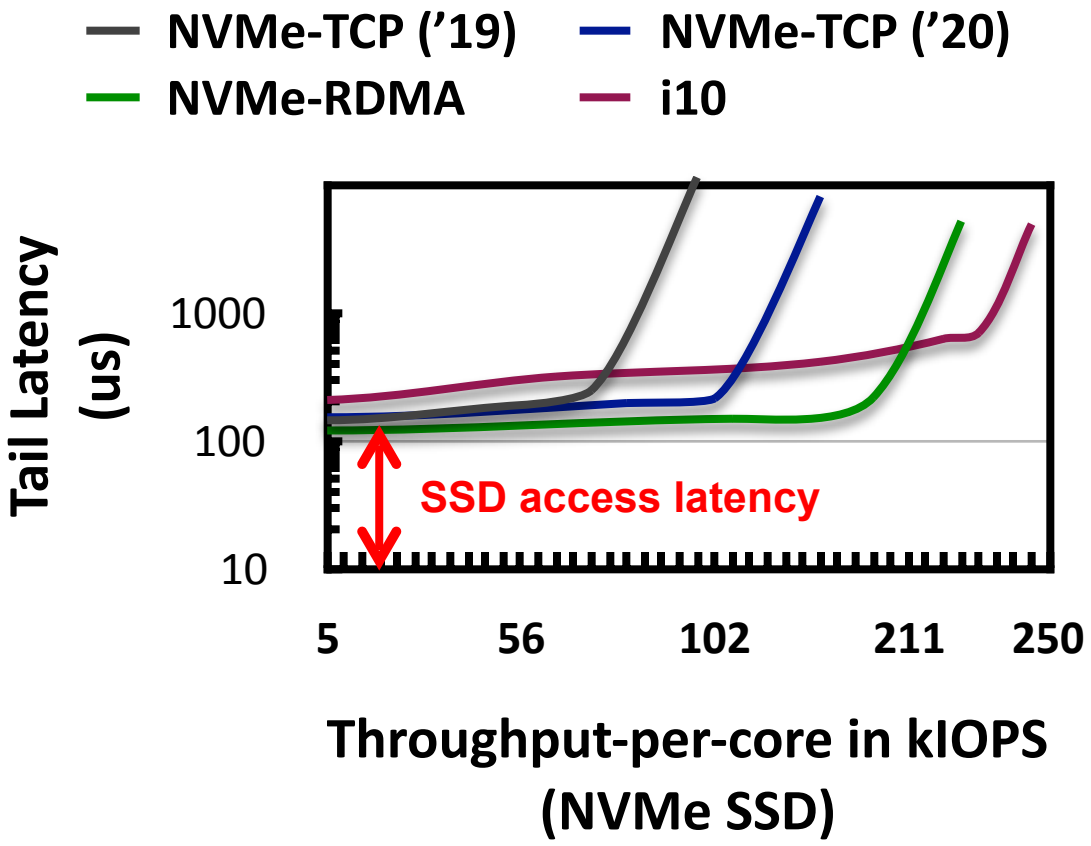
- **NVMe-TCP++ can achieve throughput-per-core similar to NVMe-RDMA**
  - Using *extremely simple* mechanisms (various forms of batching)
  - Without any specialized hardware
    - **i10-lane***: dedicated resources
    - **i10-caravans**: request and data batching
    - **Delayed doorbells**: Interrupt coalescing

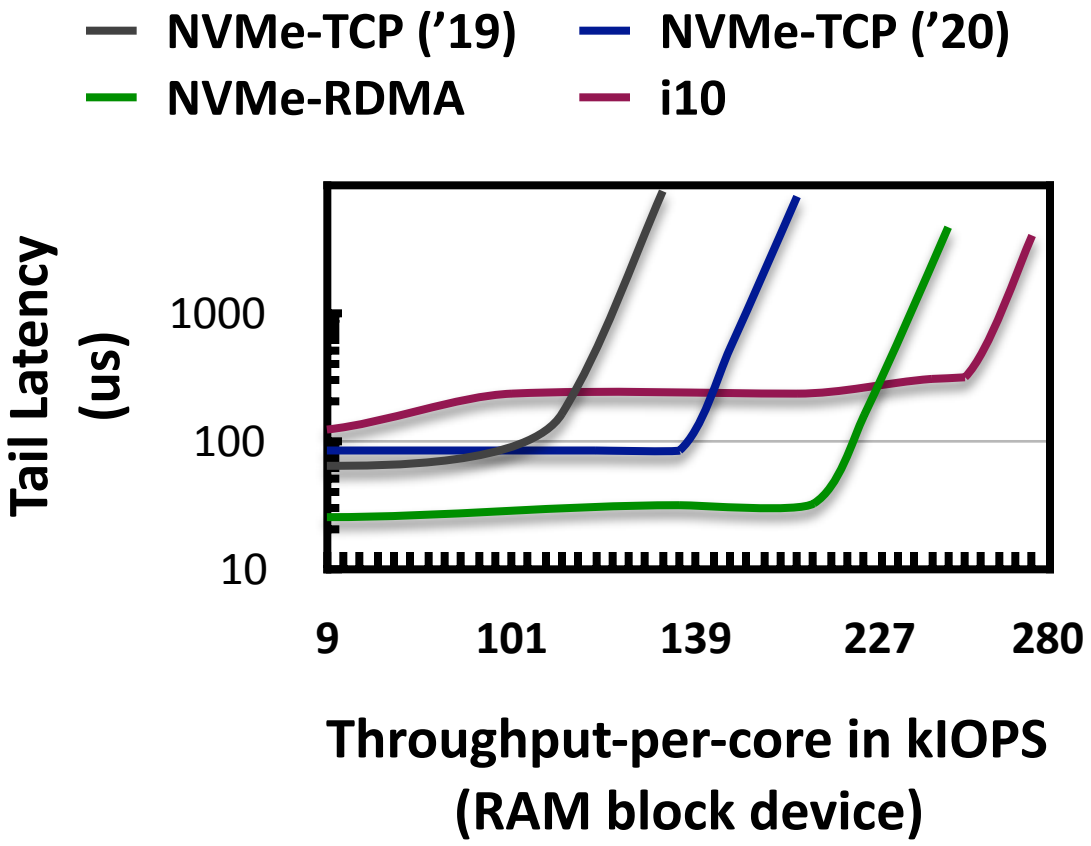


- ***Tail Latency* is ~117us higher than NVMe-over-RDMA**
  - The *real* overhead
  - Comparable to round trip times in real datacenter deployments

[*similar to NVMe-over-TCP ('20)]

i10: key take-away

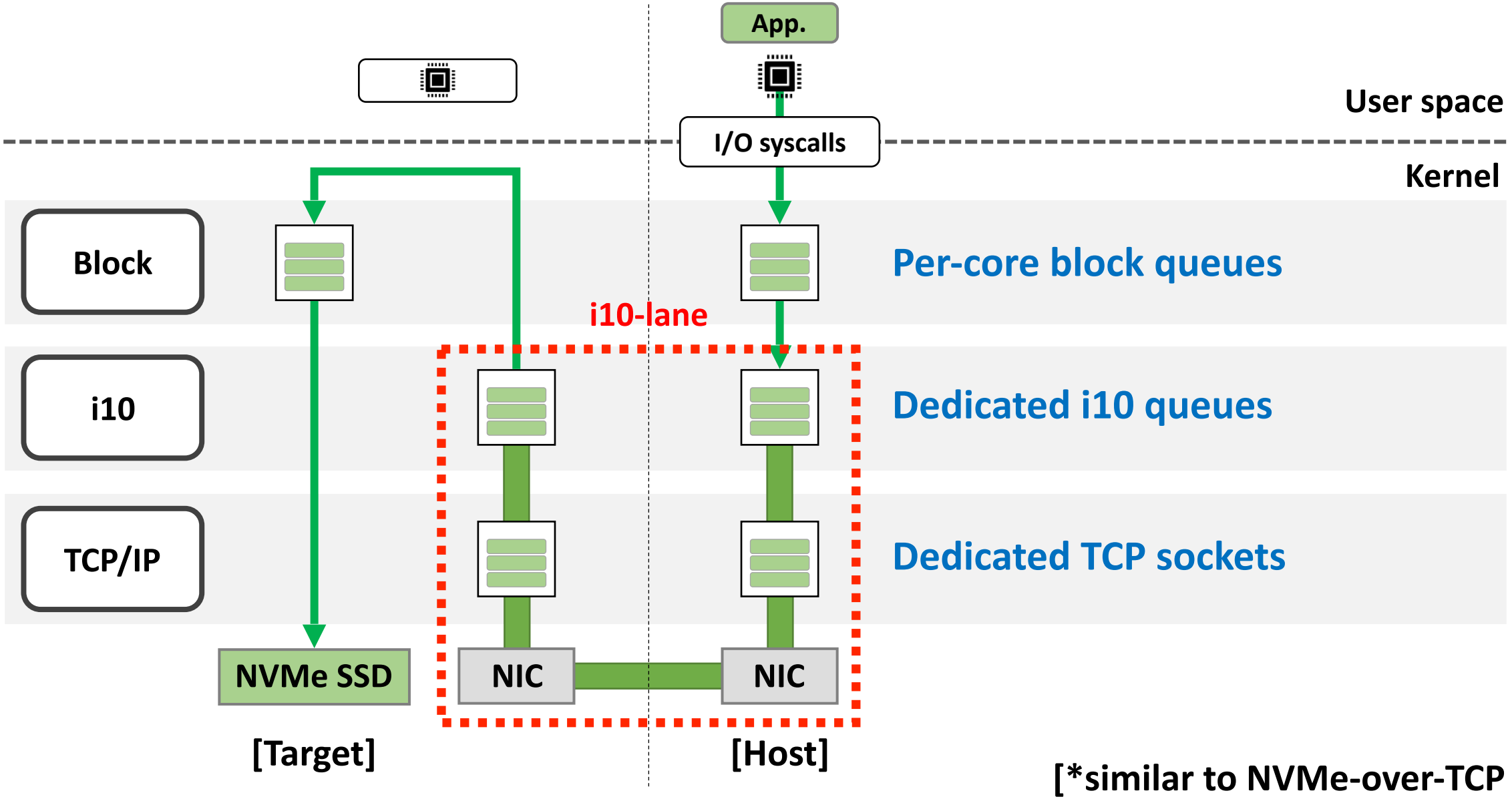




Throughput-per-core comparable to NVMe-over-RDMA can be achieved using i10

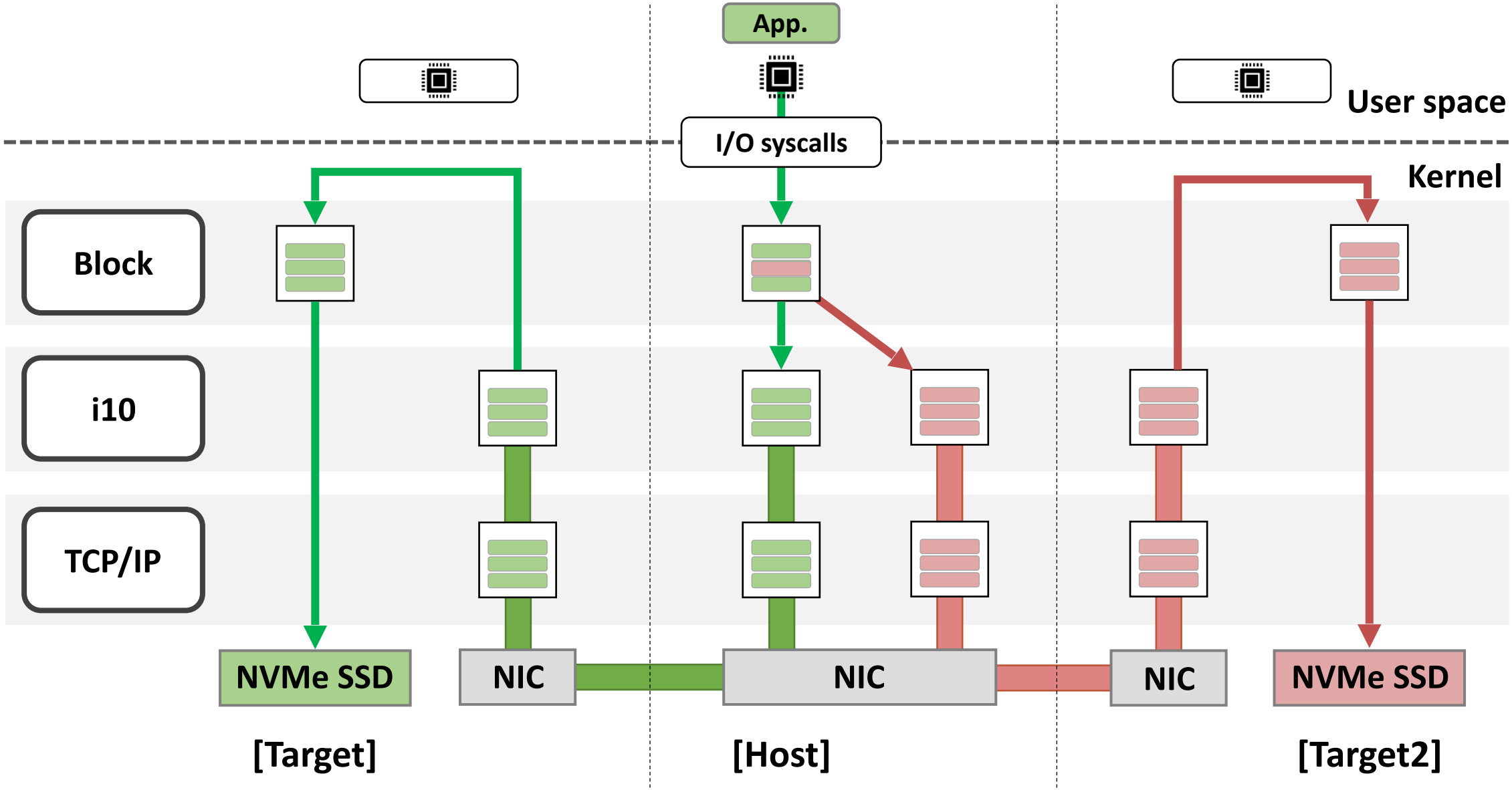
~117us higher tail latency

i10-lane*: Dedicated per-(core, target) pair pipe

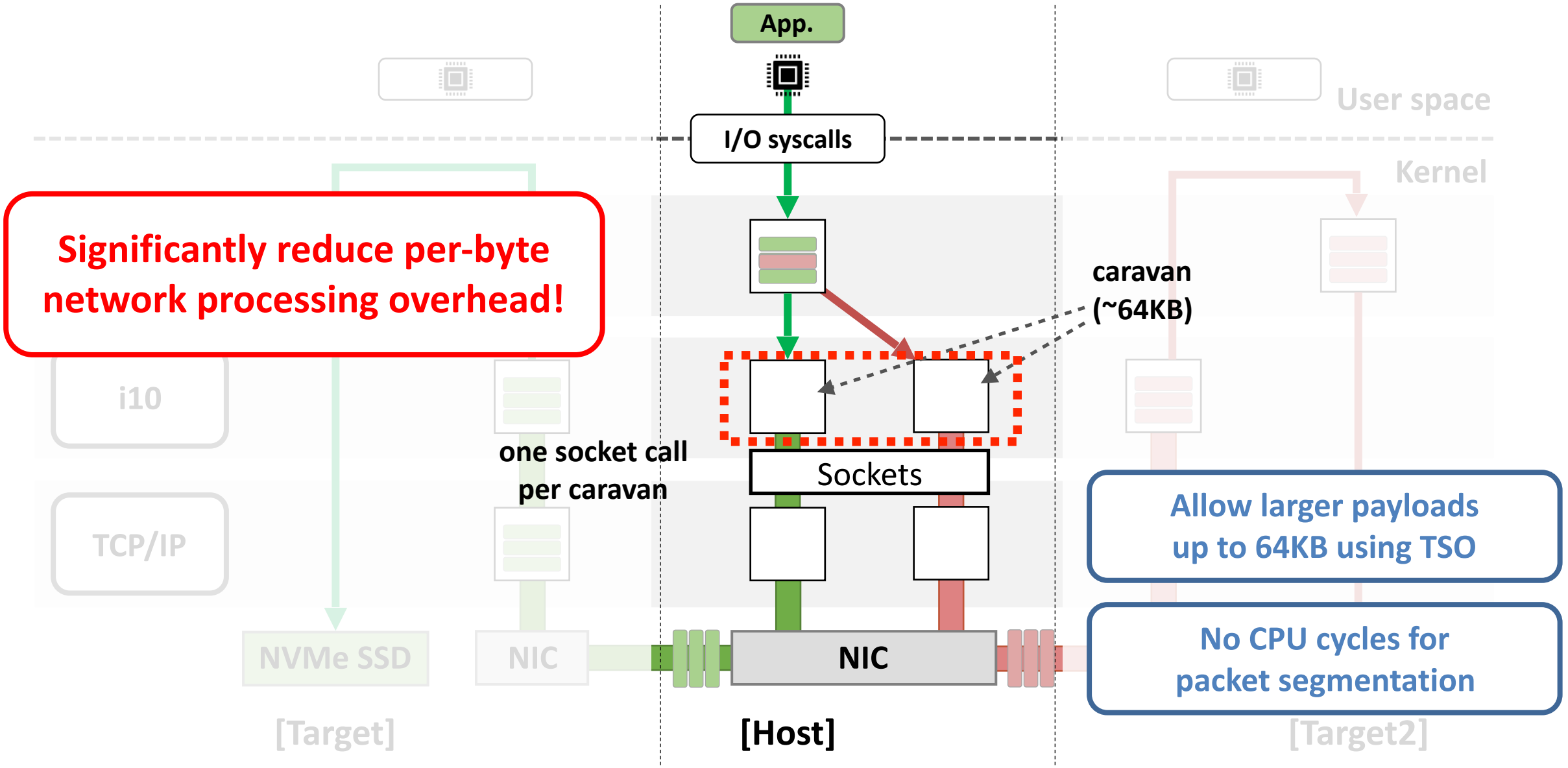


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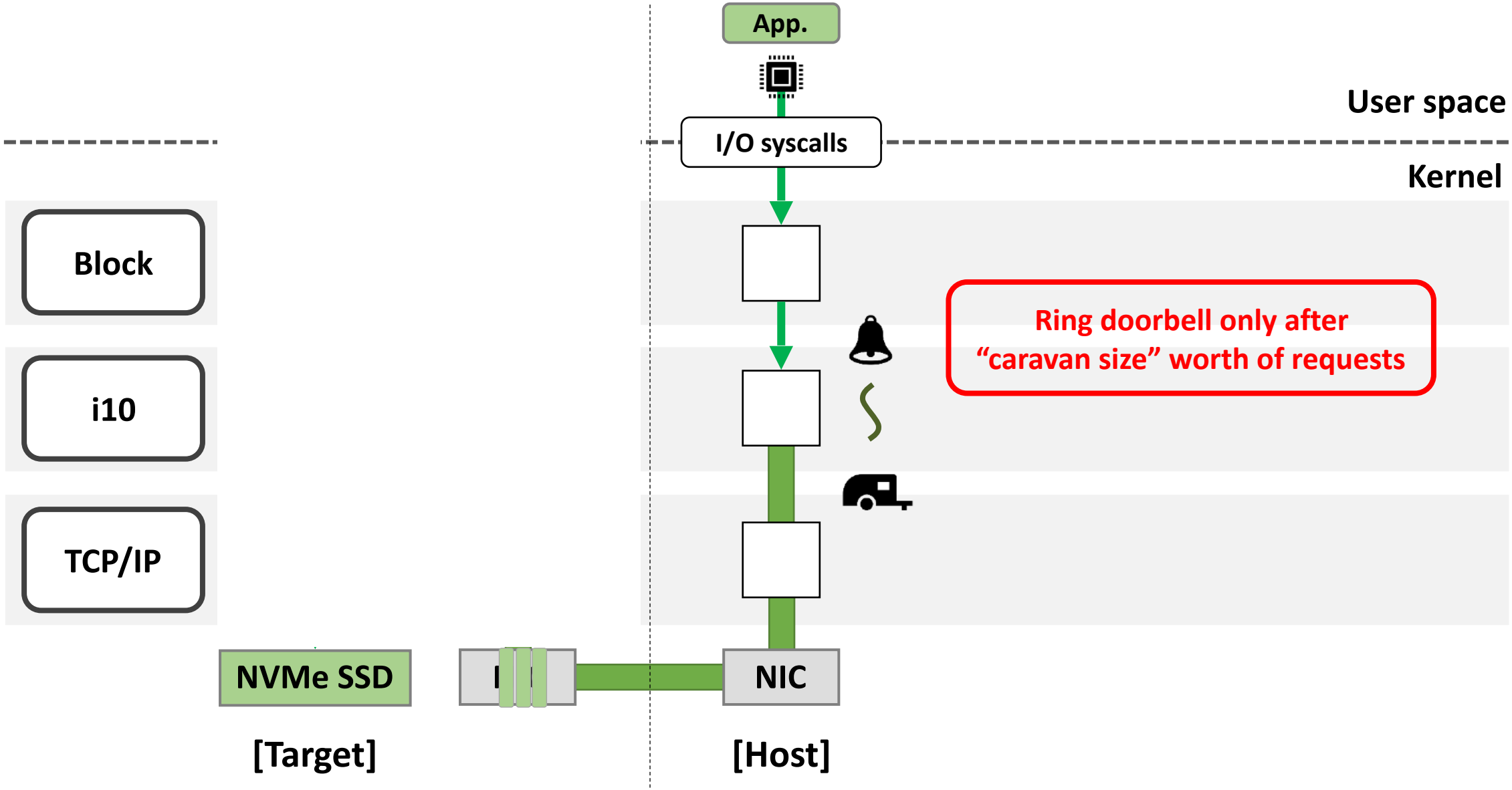
i10 Caravans: i10-lanes enable efficient batching



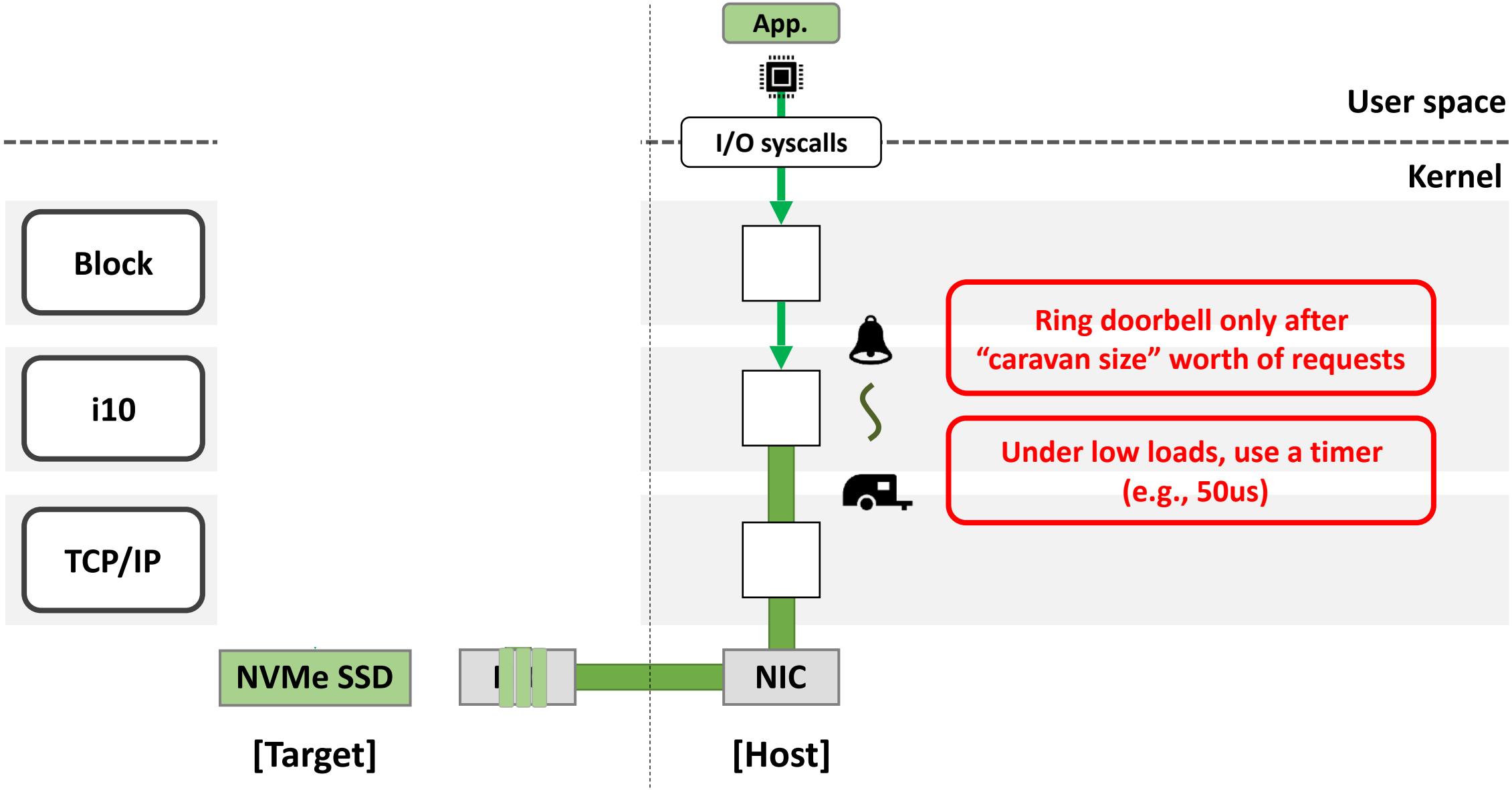
i10 Caravans: i10-lanes enable efficient batching



Delayed doorbells: Minimizing context switching



Delayed doorbells: Minimizing context switching



i10 evaluation setup

- **Two 24-core servers connected directly**
  - **100Gbps Mellanox CX-5**
  - No switches in middle — ensure bottlenecks in the kernel
    - Results will only improve with switches in the middle
- **NVMe SSD**
  - ~700k IOPS (read), ~400k IOPS (write)
  - ~100us read latency
- **No specialized hardware functionalities used in i10 evaluation**
  - For hardware and software configuration, see the paper.

i10 evaluation: how does i10 performance ……

- **… compare to NVMe-over-RDMA?**
  - Metrics of interest: throughput per core, average latency, tail latency
- **… compare to user-space stacks?**
- **… vary with different workloads, hardware and applications?**
  - read/write ratios
  - Delayed doorbell timer
  - Aggregation size
  - I/O Request sizes
  - Storage device access latency
  - Real applications
  - Number of target devices
- **… scale with number of cores?**
- **… depend on various design aspects (lanes, caravans, delayed doorbell)?**

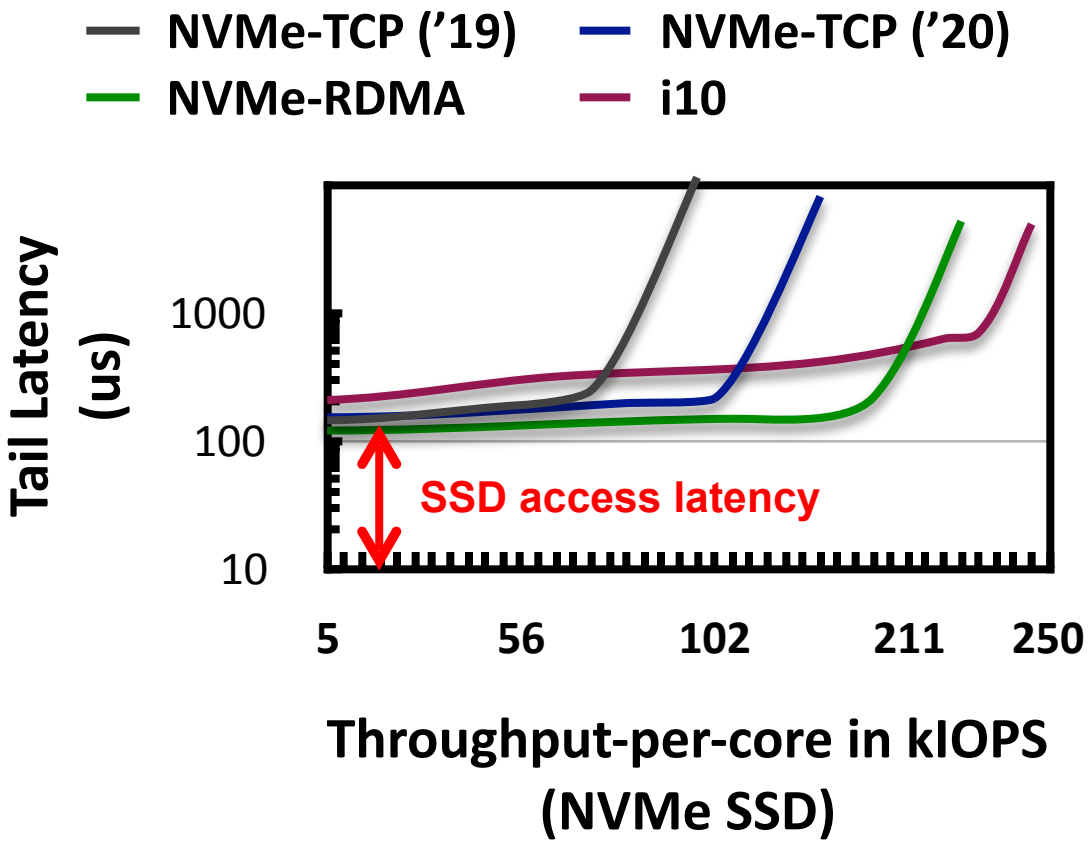
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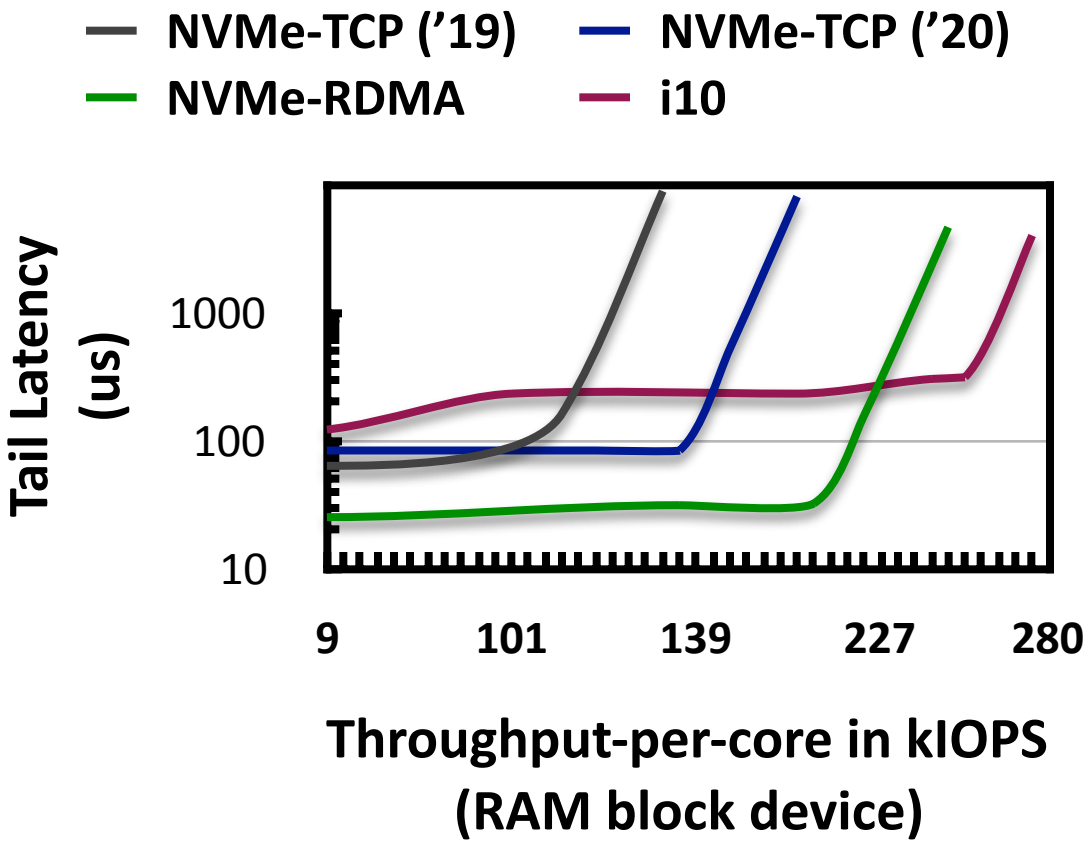
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Please see our paper!

- **… vary with different workloads, hardware and applications?**
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  - Real applications
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- **… scale with number of cores?**
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i10: Single core throughput

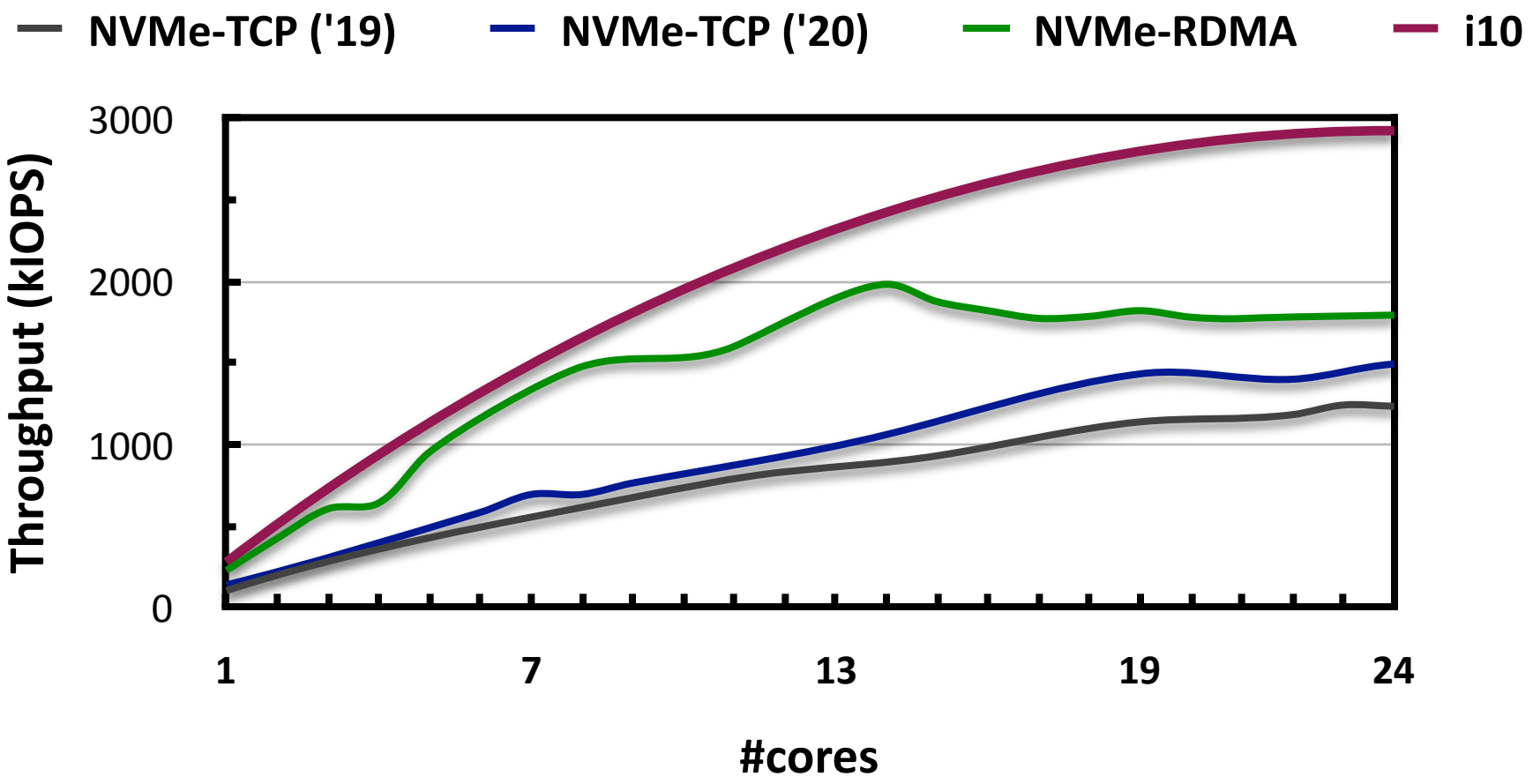




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~117us higher tail latency

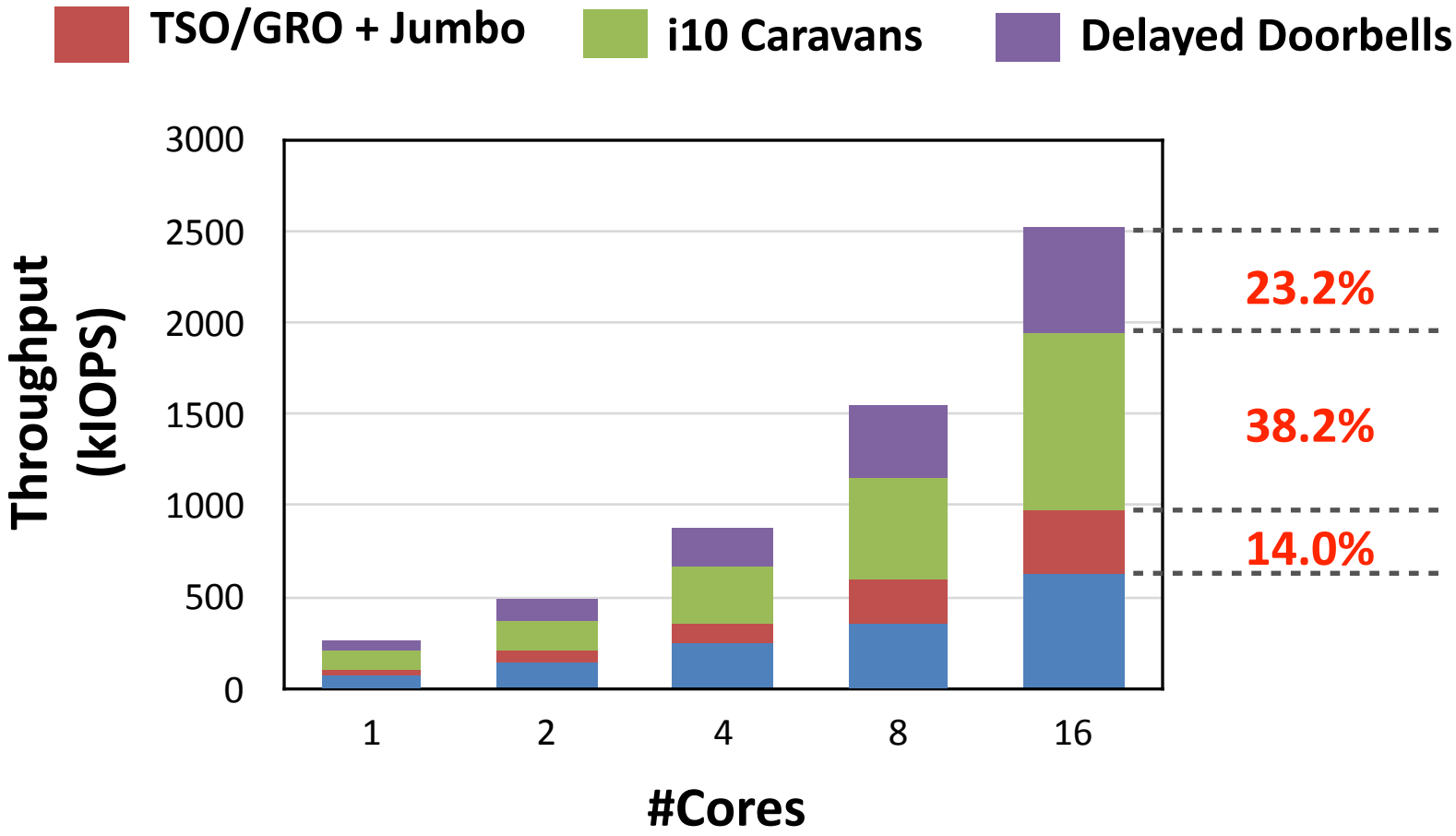
i10: Scalability with #cores (RAM block device)



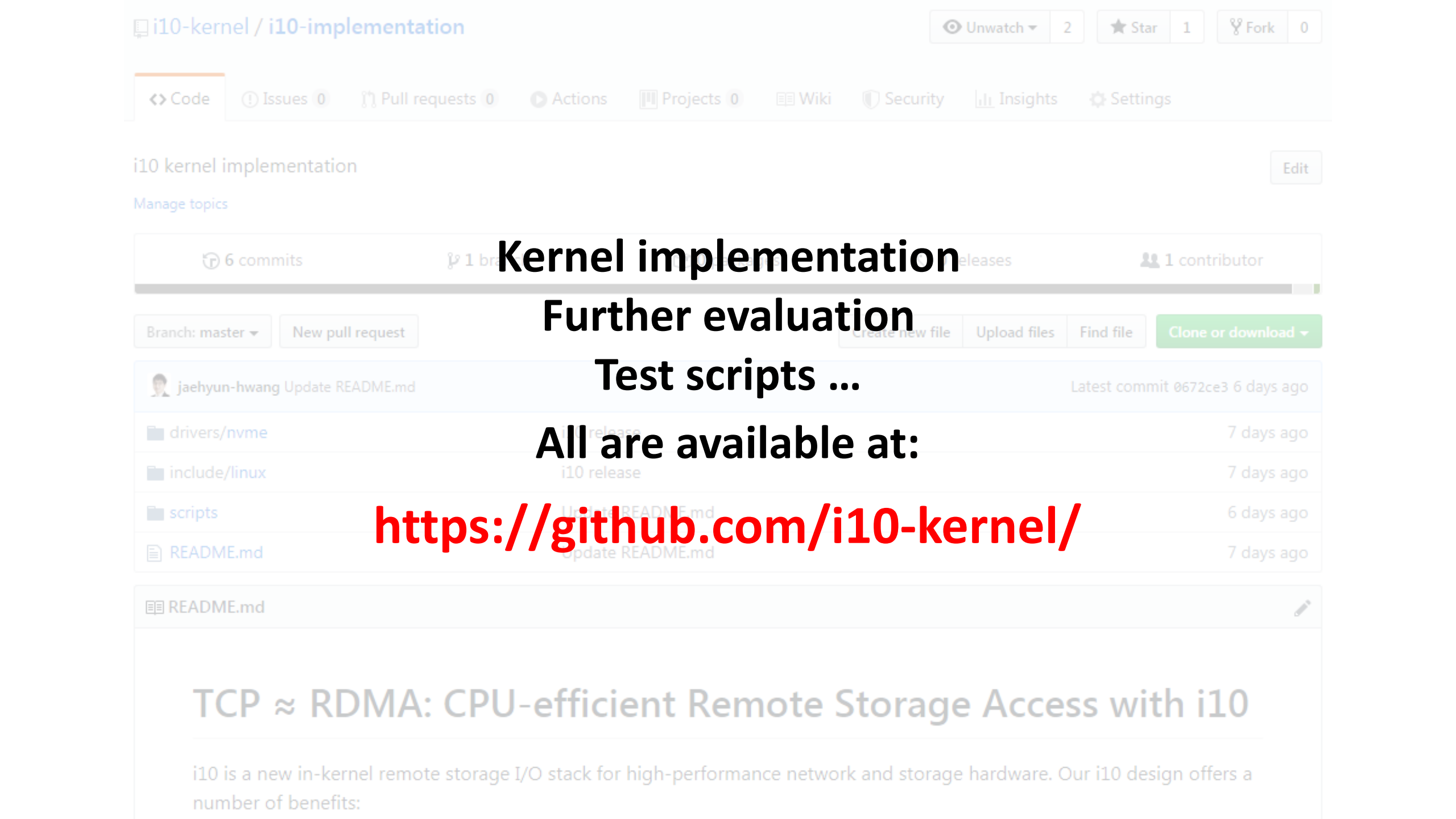
i10 scales similar to (current) NVMe-over-RDMA

For higher core count, NVMe-over-RDMA problem seems related to hardware.

i10: contribution of each component



- Each design component important for final i10 performance
- Simple batching mechanisms contribute to ~62% of the improvements



Kernel implementation

Further evaluation

Test scripts …

All are available at:

https://github.com/i10-kernel/